WORKED SOLUTION

STEP III 2002, Question 14

Probability and Statistics • Variance and Covariance

STEP III 2002, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q14

The key idea

The variance identity is just the definition of variance, expanded using linearity of expectation. The chess part is cleverer than it looks: we never compute $\mathrm{E}(WL)$ directly. Instead, we know the variance of $W + L$ from a binomial, we know $\mathrm{Var}(W)$ and $\mathrm{Var}(L)$ from binomials too, and the identity we have just proved hands us the covariance for free.

Solution

The variance identity

Write $\mu_X = \mathrm{E}(X)$ and $\mu_Y = \mathrm{E}(Y)$. We use the definitions

$$\mathrm{Var}(X) = \mathrm{E}[(X - \mu_X)^2], \qquad \mathrm{Cov}(X, Y) = \mathrm{E}[(X - \mu_X)(Y - \mu_Y)],$$

and linearity of expectation, $\mathrm{E}(aU + bV) = a\,\mathrm{E}(U) + b\,\mathrm{E}(V)$, which for discrete random variables follows by splitting the sum $\sum(au + bv)\,\mathrm{P}(U = u, V = v)$ into two sums and summing out the other variable in each.

By linearity, $\mathrm{E}(X + Y) = \mu_X + \mu_Y$, so

$$\begin{aligned}
\mathrm{Var}(X + Y) &= \mathrm{E}[((X - \mu_X) + (Y - \mu_Y))^2] \\
&= \mathrm{E}[(X - \mu_X)^2 + (Y - \mu_Y)^2 + 2(X - \mu_X)(Y - \mu_Y)] \\
&= \mathrm{E}[(X - \mu_X)^2] + \mathrm{E}[(Y - \mu_Y)^2] + 2\,\mathrm{E}[(X - \mu_X)(Y - \mu_Y)] \\
&= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y),
\end{aligned}$$

using linearity again in the third line. This is the required result. Note that no independence was needed: the identity holds for any X and Y.

Part (i)

Every game is a win, a loss or a draw, so $W + L + D = m$ whatever happens. A random variable that always takes the same value equals its mean, so its variance is $\mathrm{E}[(m-m)^2] = 0$.

Part (ii)

Call a game *decisive* if it is won or lost. Each game is decisive with probability $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$, the games are independent, and there are a fixed number m of them. So $W + L$, the number of decisive games, counts successes in m independent trials with the same success probability:

$$W + L \sim \mathrm{B}(m, \tfrac{2}{3}).$$

The correlation

In exactly the same way, W counts wins in m independent trials, each a win with probability $\frac{1}{3}$, so $W \sim \mathrm{B}(m, \frac{1}{3})$, and likewise $L \sim \mathrm{B}(m, \frac{1}{3})$. Using $\mathrm{Var} = np(1-p)$ for a binomial,

$$\mathrm{Var}(W) = \mathrm{Var}(L) = m \cdot \tfrac{1}{3} \cdot \tfrac{2}{3} = \frac{2m}{9}, \qquad \mathrm{Var}(W+L) = m \cdot \tfrac{2}{3} \cdot \tfrac{1}{3} = \frac{2m}{9}.$$

Ask yourself How do we find $\mathrm{Cov}(W, L)$ without working out $\mathrm{E}(WL)$?

Run the identity backwards. With $X = W$ and $Y = L$,

$$\frac{2m}{9} = \frac{2m}{9} + \frac{2m}{9} + 2\,\mathrm{Cov}(W, L), \qquad \text{so} \qquad \mathrm{Cov}(W, L) = -\frac{m}{9}.$$

Therefore, for $m \geq 1$ (so that the variances are non-zero),

$$\frac{\mathrm{Cov}(W, L)}{\sqrt{\mathrm{Var}(W)\,\mathrm{Var}(L)}} = \frac{-m/9}{2m/9} = -\frac{1}{2}.$$

The answer does not depend on m. It is negative, as it should be: with the number of games fixed, more wins leave less room for losses.

Part (i) gives a second route. Applying the identity twice, $\mathrm{Var}(W + L + D)$ is the sum of the three variances plus twice the sum of the three pairwise covariances. By symmetry all three covariances are equal to some c, so $0 = 3 \cdot \frac{2m}{9} + 6c$, giving $c = -\frac{m}{9}$ again.

Checks

- Take $m = 1$. Then W and L are indicators with $\mathrm{E}(W) = \mathrm{E}(L) = \frac{1}{3}$ and $WL = 0$ always, so $\mathrm{Cov}(W, L) = 0 - \frac{1}{9} = -\frac{1}{9}$, matching $-m/9$.
- The two routes (via part (ii) and via part (i)) agree.
- An exact enumeration of all 3^m outcomes for $m = 1, \ldots, 7$ gives $\mathrm{Var}(W) = 2m/9$, $\mathrm{Cov}(W, L) = -m/9$ and a correlation of exactly $-\frac{1}{2}$ each time.

Common slips

- Assuming W and L are independent because the *games* are. They are not: they share a fixed total with D.
- Taking $\operatorname{Var}(W+L) = \operatorname{Var}(W) + \operatorname{Var}(L) = 4m/9$, which forgets the covariance term and contradicts the binomial answer $2m/9$.

Beyond STEP

(W, L, D) has a *multinomial* distribution, the natural generalisation of the binomial to more than two outcomes per trial. In general, if the outcome probabilities are $p_1, \ldots, p_k$, the counts N_i and N_j have covariance $-mp_ip_j$ and correlation $-\sqrt{p_ip_j/\big((1-p_i)(1-p_j)\big)}$, which gives $-\frac{1}{2}$ here. Covariance matrices and the bilinearity of covariance are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

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